STP Implementation Options

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Brief Summary

RTGS system started its operations in July 2008. Since its inception, enrollment of the participants is growing and volume of transactions is also increasing. Payment System Department has also introduced Customer Transfers (MT103) and Multiple Credit Customer Transfer (MT102). This has further increased volume of transactions.

Currently, RTGS Participants are entering transactions in RTGS system manually using RTGS Client workplaces installed at their own premises. Same transactions they also have to re-enter in their Core Banking System through its own workplaces. Multiple re-keying is inefficient and error prone. Soon it will also be impractical to enter all these transactions manually due to increase in volume.

Therefore, it was decided to introduce STP (Straight Through Processing) to deal with the above mentioned issues. STP will electronically connect Participant Core Banking system with RTGS system using standard interfaces that are already provided with the RTGS system.

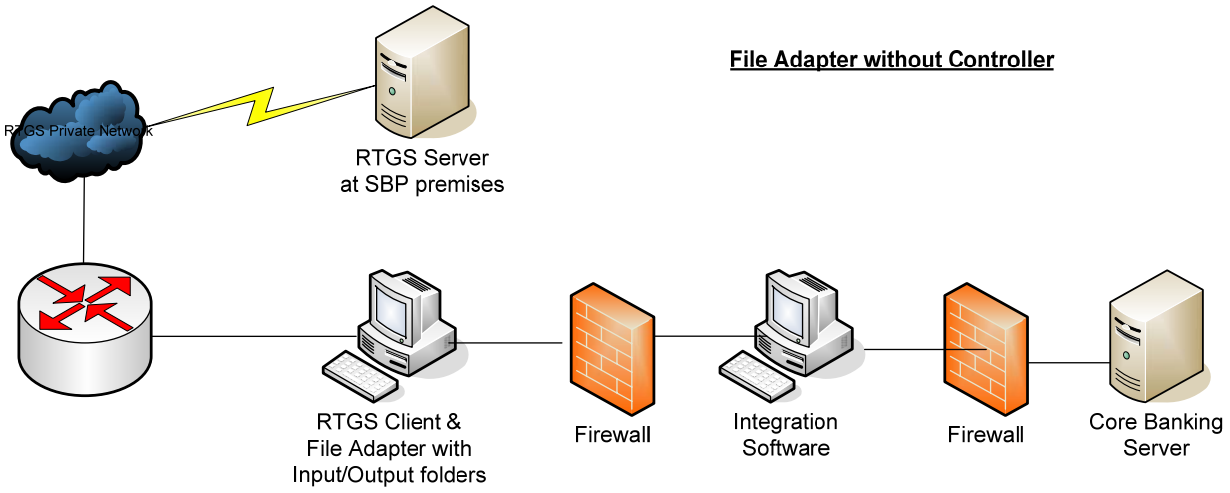
In this connection, Participants needs to develop an Integration Software that can pick up information from their Core Banking System and transfers it to RTGS system for processing and vice versa.

Following are options available for implementing STP in RTGS system.

- File Adapter (without Controller Workplace)
- Data Exchange via folders (with Controller Workplace)
- RCCC (Remote Client Communication Component)
- TMS/X

1. File Adapter without "Controller Workplace"

In this STP option, both Operator and Controller workplaces will be removed. The routines of File Adapter will work seamlessly in the background by sending and receiving payment messages through Input and Output folders to PRISM Server.



1.1 Description of Functions

1. Integration Software picks up information of an inputted transaction at Participant's Core Banking System and converts it to SWIFT like format file **(MT103 or MT202).** This file is then transferred by Integration Software to **Input** folder on RTGS Client machine where File Adapter & Simple Checker are installed.

2. Simple Checker picks up the message file from **Input** folder, applies 1st signature and then transfers it to **Intermediate** folder. 1st signature is an optional function and can be eliminated if required.

3. File adapter interface then picks up transaction from **Intermediate** folder, applies 2nd signature, and sends it to PRISM server for settlement.

4. Settlement confirmation **(MTn96)** or pending/error messages are automatically received from PRISM server in SWIFT like format by File adapter that copies it back to **Output** folder.

5. Integration Software fetches the message from Output folder, converts it to native format and provides the message to Core Banking System which finally shows it back to user. In this way user gets immediately informed about status of the transaction.

1.2 RTGS Client machine Configuration required

File Adapter & Simple Checker with RTGS libraries should be installed and client machine should be on private network and connected to PRISM Server.

1.3 Participants Software Check List

- Create files with messages according to SWIFT formats and put them in **Input** folder of File Adapter. Recognize incoming data from **Output** folder.
- Match responses to requests.
- Implement routing of received messages between Integration Software and above mentioned folders
- Take care about necessary security measures related to file exchange (via FTP or any other mechanism) e.g. managing rights on the above mentioned folders.

1.4 Identified Pros

1. Seamless operation from end to end with least human interaction.
2. Operator and Controller workplaces do not need to be installed.
3. Technicalities can be mostly handled by Participants because Integration Software is developed by their own technical team.
4. Single application Sign on is achieved in creating and sending payment messages.
5. Reconciliation efforts can also be automated at Integration Software level and hence less support requirement of SBP Technical team in case of problem at File Adapter level.

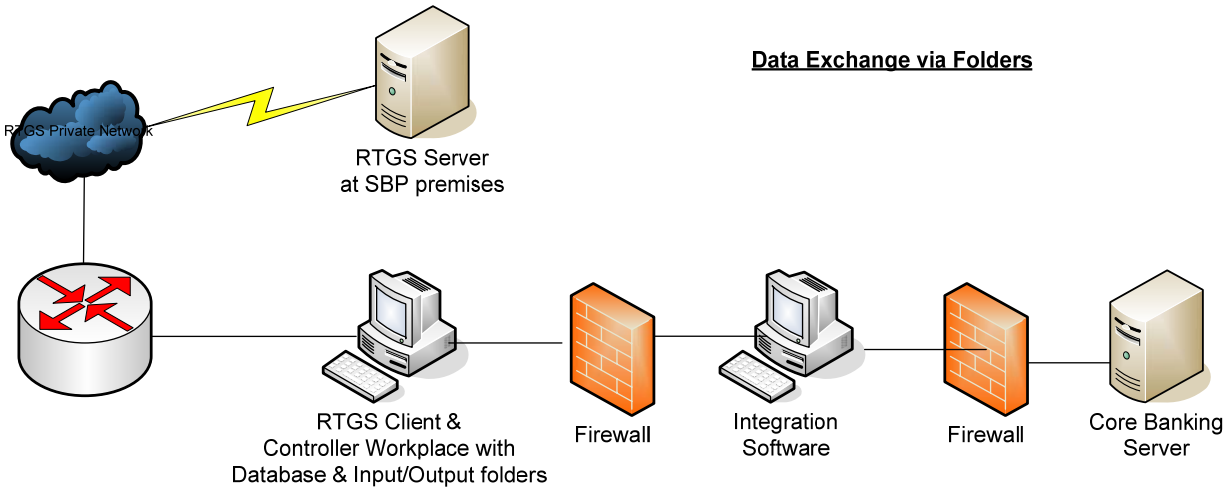
1.5 Identified Cons

1. Transaction monitoring may be difficult due to unavailability of a user interface like Controller Workplace. However, Participants can build some user interface at Integration Software level for effective transaction monitoring.
2. No local database is available for the storage of messages. Backup of messages needs to be stored in separate folders for reconciliation purpose.
3. Manual reconciliation may be difficult.
4. Handling/Management of files is required. Security controls need to be placed for that purpose.

5. It may be difficult to locate where technical problems have occurred due to unavailability of Controller Workplace

2. Data Exchange via folders With "Controller Workplace"

In this STP option, Operator Workplace will be removed although Controller Workplace still exists. Controller Workplace works by sending and receiving payment messages through Input and Output folders to PRISM Server. Copy of these messages is also stored in local database for easy reference in the future. Major difference between this and 1st option is that, manual intervention is required at Controller Workplace level for sending transactions to PRISM Server.



2.1 Description of Functions

1. Integration Software creates a SWIFT like message by using information provided by Core Banking System. This message is then transferred by Integration Software to Input folder on RTGS Client machine where Controller Workplace & Simple Checker are installed.

2. Simple Checker picks up message file from Input folder, applies 1st signature and then transfers it to Intermediate folder. 1st signature is an optional function and can be eliminated if required.

3. Controller Workplace then picks up transaction from Intermediate folder stores it in Local Database and shows it to Authorizer for approval.

4. Authorizer of Participant bank, after review & verification, sends this transaction to PRISM server via Controller Workplace. Signature is also applied at this level.

5. If Authorizer finds message credentials incorrect, he can send it back to the Integration Software. It is done by copying the message back to Output folder via Controller Workplace

without sending it to PRISM Server. Integration Software then fetches Output folder and provides the message to the Core Banking System which shows it back to user for correction.

6. When transaction is sent to PRISM server, settlement confirmation or 'Error Message' is automatically received from PRISM server by Controller Workplace that copies it back to Output folder as well as local database.
7. Integration Software fetches **Output** folder and provides message to the Core Banking System which shows it back to user for information/correction.

2.2 RTGS Client machine Configuration required

RTGS libraries should be installed. Local database in MS Access is installed with Controller Workplace and client machine is connected to PRISM server.

2.3 Participants Check List

- Create files with messages according to SWIFT like formats and then put them in Input folder of Controller Workplace. Recognize incoming data from Output folder.
- Setting options of Controller Workplace so that it can automatically import messages from Input folder
- Setting options of Controller Workplace so that it can export messages to Output folder
- Match responses to requests.
- Implement routing of received messages between Core Banking system and above mentioned folders.
- Taking care about necessary security measures related to file exchange (via FTP or any other mechanism) e.g. managing rights on the above mentioned folders.

2.4 Identified Pros

1. Operator Workplace doesn't need to be installed. Payment messages come directly in Input folder from Core Banking system through Integration Software developed by Participant's technical team.
2. Technicalities can be mostly handled by Participants because Integration Software is developed by their own technical team.
3. Controller workplace is available for sending and viewing transactions. This saves development efforts required for some monitoring application as mentioned in 1st STP option. Local database will also be available for storing messages locally.
4. Reconciliation will be easier due to availability of Controller Workplace.

5. It may be easier to locate where technical problems have occurred, i.e. at Participant end or at PRISM end due to availability of Controller Workplace.

2.5 Identified Cons

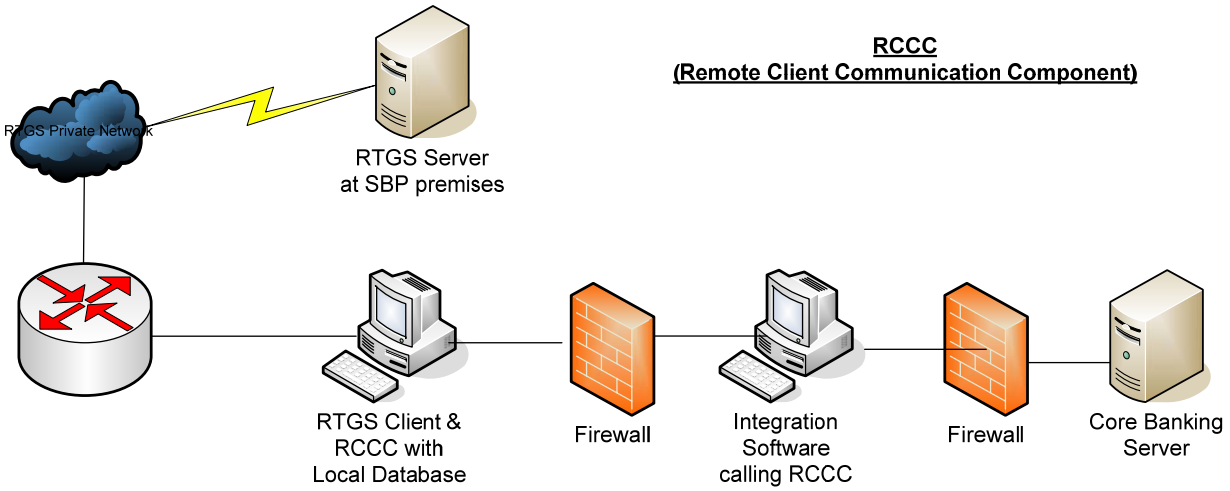
6. Although re-keying is eliminated, but manual intervention of Authorizer is still required for sending payment message to PRISM Server using Controller Workplace

7. Increased support requirement of SBP Technical team in case of problem in Controller Workplace as compare to 1st option.

8. Handling/Management of files will still be required.

3. RCCC (Remote Client Communication Component)

In this STP option, both Operator and Controller Workplaces don't exist. RCCC is basically an API developed by CMA (RTGS vendor) specifically for the purpose of STP. This API provides many functions that may be required by Participants' technical teams for implementing STP.



3.1 Description of Functions

1. Integration Software interacts with RCCC by using DCOM and RPC. By using RCCC, Integration Software can directly send/receive messages to PRISM server. Information required is passed via function parameters to PRISM Server.

2. Settlement confirmations/error messages can be easily received by the Integration Software from PRISM server as a return parameter of RCCC function.

3. It is preferred to use local database for making a copy of all the messages exchange in between Integration Software and PRISM server. It makes reconciliation process easier.

4. It is also preferred to make some user interface in Integration Software that shows message exchange in between Integration Software and PRISM server. It also makes reconciliation process easier.

3.2 RTGS Client machine Configuration required

RCCC installed and connected to PRISM server. RTGS libraries installed. Local database (if used) installed.

3.3 Participants Check List

- Embed RCCC library interface into Participant's Integration Software.
- Provide message fields and parameters to RCCC as per PRISM server formats.
- Match responses to requests.
- Recognize incoming data as function return parameter.
- Implement routing of received messages between Core Banking System and PRISM Server

3.4 Identified Pros

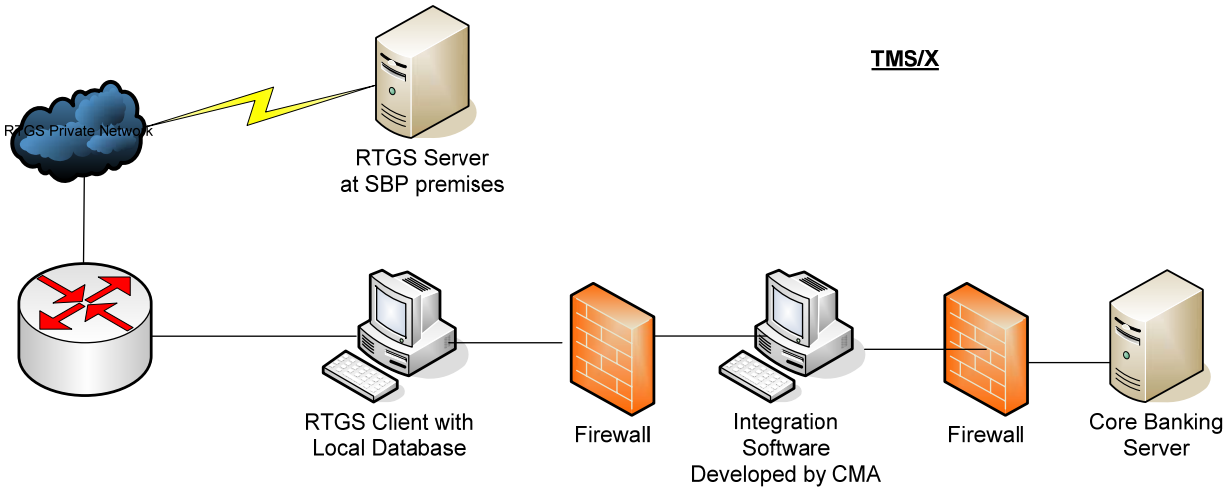
1. Both Operator and Controller workplaces don't need to be installed.
2. Technicalities can be handled by Participants' technical teams because all interfacing is developed by them.
3. More control over interfaces.
4. Seamless working will be achieved.
5. Single application Sign on is achieved in creating and sending payment messages.
6. Local database can be created (if required) and it will make reconciliation easier.
7. More secure than file based options

3.5 Identified Cons

1. Integration Software with transaction monitoring facility needs to be developed.
2. More technicalities involved. Integration Software will be more complex.
3. It may be difficult to locate where technical problems have occurred due to unavailability of Controller Workplace

4. Transaction Management System TMS/X

In this STP option, CMA (Vendor of RTGS) will provide a complete solution. It may be a hybrid solution and use any possible combination of the above mentioned options depending upon Participant's requirements. Integration Software may also be developed by CMA.



4.1 Description of Functions

1. All integration will be done by CMA with respect to sending and receiving messages from Participant's Core Banking system and PRISM server. Any of the above mentioned options or their combination may be used.

2. It is preferred to use local database for making a copy of all the messages exchange in between Integration Software and PRISM server. It makes reconciliation process easier.

3. It is also preferred to make some user interface in Integration Software that shows message exchange in between Integration Software and PRISM server. It also makes reconciliation process easier.

4.2 RTGS Client machine Configuration required

RTGS libraries should be installed. Local database in MS Access should be installed. Other installations depend upon Participant's requirements and type of solution recommended by CMA.

4.3 Participants' Check List

- Participant shall contact separately to CMA for the development of complete solution including Integration Software.
- Participant will explain the interfaces with Core Banking system to CMA for development

4.4 Identified Pros

1. CMA as a vendor of RTGS can better understand RTGS interfaces. They can design & develop high quality Integration Software as per requirements of Participants.
2. Limited development efforts will be required by Participant's technical teams because all interfaces will be developed by CMA.
3. Seamless working & single Sign On can be achieved if required by Participant.
4. Support/Maintenance can also be provided by CMA if required by Participant.

4.5 Identified Cons

1. Technical dependency on CMA.
2. More expensive than other options.
3. Separate contracting with CMA by participant is required.